

User Retention & Business Decision Impact Analysis

Data Source Details: Brazilian E-Commerce Public Dataset by Olist

The dataset used in this project is the Brazilian E-Commerce Public Dataset provided by Olist, containing detailed order-level information from 2016 to 2018. [Brazilian E-Commerce Public Dataset by Olist](#)

Olist is a Brazilian e-commerce company that acts as a marketplace integrator, enabling small and medium-sized businesses to sell their products across multiple online marketplaces through a single unified platform. It simplifies operations by providing services such as product listing, order management, payment handling, and logistics coordination.

This dataset is anonymised, meaning that all sensitive business information has been modified to ensure privacy. References to companies and partners have been replaced with fictional names (such as Game of Thrones houses), while preserving the overall structure and analytical usefulness of the data.

The dataset includes multiple related tables covering customers, orders, payments, products, reviews, sellers, and geolocation, making it suitable for analysing customer behaviour, retention patterns, and business performance.

The dataset includes the following key tables:

- Customers: customer details and location
- Orders: order status and timestamps
- Order Items: product-level transaction details
- Payments: payment values and types
- Products: product attributes and categories
- Sellers: seller information
- Reviews: customer feedback
- Geolocation: zip code, city, and state data
- Product category names were translated using the official product translation table.

Data Import Process:

- The dataset was loaded into PostgreSQL using Python.
- Used pandas and SQLAlchemy to read CSV files
- Connected to PostgreSQL using psycopg2
- Converted all column names to lowercase for consistency
- Loaded each file into the database as raw tables using `to_sql()`
- All tables were stored with the prefix `raw_` (e.g., `raw_orders`, `raw_customers`).

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Data Cleaning and Preprocessing

Data cleaning was performed in a structured manner to ensure consistency, accuracy, and completeness of the dataset before analysis.

Handling Missing Values

Missing values were identified and treated based on the nature of each column.

i) Missing values were found in: Product Category

A default category 'Unknown' was introduced in the translation table

All NULL values were replaced with '**Unknown**'

ii) Missing values were found in: Product table

- product_weight_g
- product_width_cm
- product_height_cm
- product_length_cm

These were handled by:

Replacing NULL values with the average (AVG) of each column

Geolocation Data Challenges

The geolocation data contained **errors in city names** due to special characters and was **missing some zip codes** that were present in the customer and seller tables.

- Missing zip codes in:
- Customers table (157 records)
- Sellers table (7 records)

1. Text Standardisation

- To make the location data consistent, city names were cleaned by removing special characters and converting all text to lowercase. The unaccent function was also applied to ensure all city names followed a standard format.
- A new column city_clean was created to store cleaned city names.

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2. Master Zip Code Creation

To resolve missing and inconsistent location coverage:

- Zip codes were combined from:
- Geolocation
- Customers
- Sellers

A UNION operation was used to create a master list of unique zip codes and this ensured complete coverage of all locations across the dataset.

3. Geolocation Aggregation

To ensure a single, consistent record per zip code, latitude and longitude were aggregated using AVG(), while city and state were standardised using MAX() on cleaned values

4. Created Geolocation Dimension

The final dim_geolocation table was created by joining the master zip code table with aggregated data to ensure one record per zip code, complete mapping for customers and sellers, and clean, consistent geographical information.

5. Handling Missing Geolocation Values

After processing:

- Missing latitude and longitude (162 records) - replaced with 0
- Missing city values - replaced with 'Unknown'
- Missing state values - replaced with 'N/A'

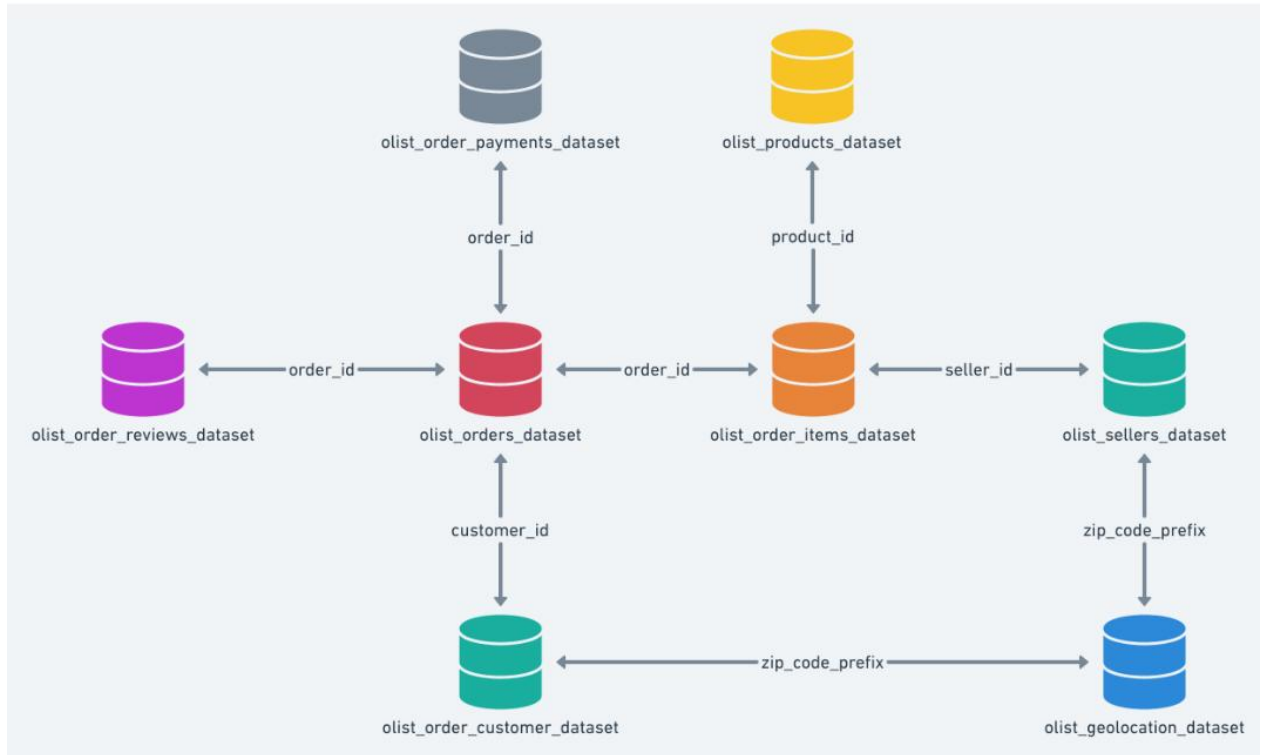
Keys & Indexing

- Primary keys created for all tables
- Foreign keys established
- Indexes created for:
 - order_id
 - product_id
 - customer_id etc.

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Data Schema:

The data is structured into multiple specialized datasets to ensure modularity and organizational clarity. This schema defines the core relational architecture, mapping how orders serve as the central link between customers, products, and logistics providers.



Project Overview

This project titled **“User Retention & Business Decision Impact Analysis”** focuses on understanding customer behaviour, retention patterns, revenue contribution, and operational impact using e-commerce transaction data.

The main objective of the project is to analyse how customers behave after their first purchase, identify factors influencing repeat purchases, and evaluate how business operations such as delivery performance, product categories, and payment methods affect long-term customer engagement.

The analysis is built using 20 SQL business questions and is supported by dashboards covering customer retention, cohort behaviour, product performance, and logistics impact. These analyses help in understanding how different business factors contribute to customer retention and overall revenue performance.

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Overall Business Problem

- Lack of clear understanding of customer retention and repeat purchase behaviour, making it difficult to identify loyal vs one-time customers
- Inability to properly analyse cohort patterns and customer lifecycle trends over time, which limits understanding of how customers return and engage
- Limited visibility into customer segmentation (one-time vs repeat) and their contribution to overall revenue, affecting targeting strategies
- Difficulty in identifying high-value customers and potential churn risks, leading to missed opportunities to retain valuable users
- No clear estimation of revenue impact due to the loss of top customers, making it hard to plan revenue growth strategies
- Unclear impact of delivery performance (on-time vs late) on customer retention, affecting service improvement decisions
- Limited insights into how different product categories influence repeat purchases and customer behaviour
- Lack of clarity on active customer trends and engagement over time, making it difficult to track business growth
- Difficulty in making better data-driven decisions to improve customer retention and overall revenue performance

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Business Questions:

Q1. Quantify customer retention by segmenting delivered-order customers into one-time shoppers and repeat loyalists. Determine the population distribution and relative segment ratios to establish a baseline for brand loyalty and its potential impact on customer acquisition costs.

Insights: 97% customers are one-time shoppers, showing extremely low repeat purchase behaviour. Only 3% customers return, indicating weak customer loyalty and poor retention strength.

Recommendations: Focus on improving first purchase experience (delivery, product satisfaction, support) to increase repeat rate. Introduce retention strategies like discounts, reminders, or loyalty rewards to convert one-time buyers into repeat customers.

Q2. Identify each customer's acquisition cohort by determining the month of their first delivered purchase. This establishes the foundation for cohort-based retention analysis.

Insights: 93,358 customers have been assigned a cohort based on their first delivered purchase month.

Recommendations: Use cohort field as the base for retention and lifecycle analysis.

Q3. Define and measure active customers by calculating the number of unique customers who made a purchase within recent time windows (e.g., last 30 days vs 60 days). This helps assess current engagement and platform activity levels.

Insight: Active customers in the last 30 days are 0, meaning there is no recent customer activity. In the last 60 days, some customers are active, but they still stop returning after a short time.

Recommendation: Immediately run reactivation campaigns (emails, WhatsApp offers, discounts) to bring back inactive customers. Improve post-purchase engagement so customers return within 30 days instead of going inactive.

Q4. Analyze monthly active customer trends across different years to determine whether observed fluctuations are driven by seasonal patterns or indicate abnormal changes in customer behaviour.

Insight: Customer activity increases strongly over time, showing steady business growth from 2017 to 2018. There is a clear peak in November 2017, which is likely due to seasonal shopping events like Black Friday sales.

Recommendation : Plan major marketing campaigns around high-performing months like November to maximize sales impact. Investigate low months (early 2016 and mid-2017) and improve marketing efforts during slow periods.

Q5. Evaluate the relationship between delivery performance (on-time vs late) and customer retention by analyzing repeat purchase behaviour across different delivery experience segments.

Insight : On-time deliveries have a higher repeat purchase rate (3.04%) compared to late deliveries (2.52%). This shows that delivery performance has a direct impact on customer retention.

Recommendation : Improve logistics to reduce late deliveries and ensure faster, reliable shipping. Focus especially on first-order delivery experience to increase customer repeat purchases.

Q6. Weekend vs Weekday order distribution analysis

Insight: Weekday orders are much higher (74288) compared to weekend orders (22190).

Recommendations: Focus operations, staffing, and campaigns more on weekdays, since most orders are placed during this period.

Q7. Calculate revenue loss if 5% of top 20% high-value customers don't return next quarter.

Insight: Revenue loss estimate from top 20% high-value customers is 412,733.28. Top 20% customers contribute a significant portion of total revenue.

Recommendation: Focus on maintaining high-value customer contribution to revenue. Track high-value customer revenue contribution regularly.

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Q8. Which product categories have the highest average time between repeat purchases?

Insight: fashion_bags_accessories (13.36 days) and musical_instruments (12.75 days) have the highest average time between repeat purchases. electronics (1.95 days) and furniture_decor (1.67 days) have low average time between repeat purchases.

Recommendation: Focus on categories with high repeat purchase time to improve customer return speed. Maintain performance in categories with low repeat purchase time as they show faster repeat buying.

Q9. How much revenue does an average customer generate?

Insight: Average revenue per customer is 165.20. Each customer generates around this amount on average.

Recommendation: Use this value to track customer value over time and focus on increasing average customer spend.

Q10. How is revenue distributed across our customer base by quintile?

Insight: The highest revenue concentration is in Quintile 1, which contributes 53.53% of total revenue and revenue contribution decreases steadily from Quintile 1 to Quintile 5, showing uneven distribution across customers.

Recommendation : To analyse purchase patterns, product categories, and payment behaviour of Quintile 1 customers and apply similar strategies to other segments. Use targeted offers, discounts, and personalized marketing to increase spending from lower quintile customers.

Q11. Identify high-value customers who have not made a purchase in the last 90 days, highlighting potential churn risk among top revenue customers

Insight: 14,915 high-value customers have not made recent purchases, indicating a large inactive premium customer base.

Recommendation: Target these inactive high-value customers with re-engagement campaigns like discounts, personalized offers, and reminder emails. Monitor high-value customer activity regularly to reduce churn and improve retention.

Q12. Evaluate how customer review ratings influence spending behaviour by comparing average order values across different rating levels.

Insight: Customers with 1-star rating show the highest average order value (186.39) compared to all other rating groups. Higher ratings (4–5 stars) show lower average order values (147–149), while lower ratings also vary but remain lower than 1-star group.

Recommendation: The 1-star group has the highest average order value (186.39), so these higher-spend orders should be prioritised for issue tracking (delivery, product mismatch, service errors) at order level. Since all rating groups (1–5) still show similar average values (145–186 range), introduce rating-based monitoring on high-value orders rather than treating all low ratings equally.

Q13. Find "Loyalists"—customers whose every subsequent order amount is greater than their very first order amount.

Insight: These customers show rapid trust by frequently doubling their spend on second orders.

Recommendation: Offer rewards after the first purchase to encourage these customers to spend more.

Q14. Determine each customer's "birth month" based on their first delivered purchase to support cohort segmentation and lifecycle analysis.

Insight: 93,358 customers have been assigned a cohort based on their first delivered purchase month.

Recommendation: Track customer cohort-wise retention to compare how different acquisition months perform in repeat purchases. Use this cohort structure to identify which months bring more loyal customers and which months need improvement.

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Q15. Build a cohort table showing how many users from Jan 2017 returned in Feb, March, etc. (Calculated as Month 0, Month 1, etc.)

Insight: Most customers stop interacting with the brand immediately after their first purchase.

Recommendation: Review the post-purchase experience to find better ways to keep users interested.

Q16. Compare the average total spend (LTV) of users who gave and Compare LTV of customers based on first order review score (1-star vs 5-star)

Insight: Customers who gave a 1-star review on their first order have a higher average lifetime value (199.13) compared to customers who gave a 5-star review (161.87). This shows that higher spending customers are not always more satisfied.

Recommendation: Investigate high-spending customers who gave 1-star reviews and fix issues in their purchase experience. Improve on service quality for high-value orders to reduce dissatisfaction and improve retention.

Q17. Compare average freight costs and repeat purchase rates across key states (SP vs RJ) to evaluate whether higher logistics costs are associated with lower customer retention.

Insight: RJ shows higher average freight cost (23.95) compared to SP (17.33). Repeat user percentage is almost similar in both states, RJ at 3.28% and SP at 3.13%, even though SP has a much larger customer base.

Recommendation: Track why RJ has higher freight cost (23.95) compared to SP and monitor its impact on cost efficiency. Monitor repeat purchase levels in both states to maintain consistent customer return rates despite freight differences.

Q18. At what time of day do customers place most orders?

Insight: Highest orders are at 16:00 (6476) and strong demand also at 11:00 and 14:00. Very lowest orders are at 05:00 (182) and 04:00 (203).

Recommendation: Focus on business activity during peak hours like 11:00–16:00, and use early morning hours (04:00–05:00) for maintenance or low-activity tasks.

Q19. What is the average time in hours for an order to be approved

Insight: The average order approval time is 10.42 hours from purchase to approval.

Recommendation: Check on reducing approval delays by reviewing internal approval process flow to improve overall order processing speed.

Q20. Which are the top 10 revenue-generating cities, and how does payment type (credit card, debit card, UPI, etc.) contribute to revenue distribution within those cities?

Insight: Credit card is the dominant payment method across all top cities and generates the highest revenue contribution. Boleto is consistently the second most used payment method, while debit card and voucher contribute minimally across all cities.

Recommendations: Maintain the current payment structure since it already reflects stable customer behaviour across cities. Use this breakdown as a baseline to track how payment preferences change over time in different regions.

User Retention & Business Decision Impact Analysis

Customer Retention Overview Dashboard



Insight

The current state of customer loyalty is a major challenge, with a 97.00% Churn Rate and only 2801 Repeat Customers out of a total of 93K. While there are 6,760 Active Customers (30D), most people only buy once, resulting in a very low Avg Orders Per User of 1.03. The Churn Rate by Month stays extremely high throughout the year, peaking at 95.51% in March and 95.76% in August. Interestingly, retention varies by product; the Bed Bath Table category has the highest Repeat Rate at 6.42%, while Watches Gifts is much lower at 3.67%. Additionally, the Total Customers by Quarter shows a significant drop in the fourth quarter, falling to 16.5K compared to the 27.0K peak in Q2.

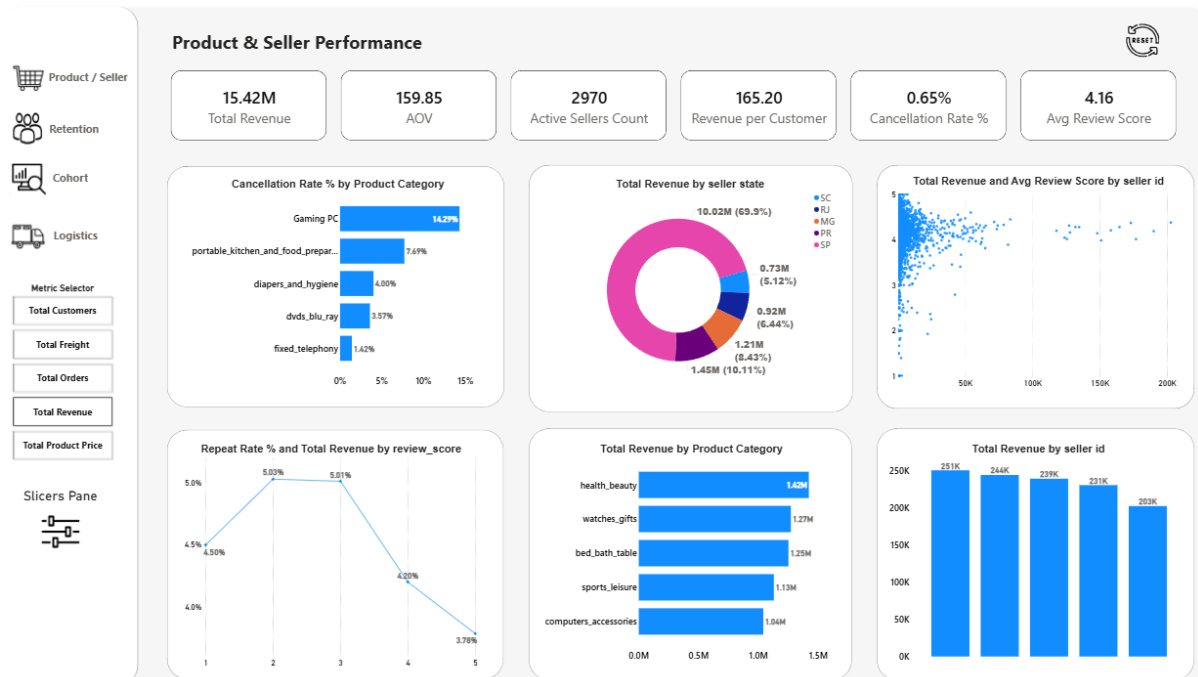
Recommendation

A massive effort is needed to lower the 97.00% Churn Rate by focusing on the 81,472 customers who have only placed one order. Since the Bed Bath Table category shows the best Repeat Rate (6.42%), successful strategies from that department should be applied to lower-performing areas like Watches Gifts. To address the 95.76% churn peak in August, targeted re-engagement campaigns or early "end-of-summer" loyalty rewards should be launched to keep those users active. Finally, with the sharp drop in new customers during Q4 (16.5K), a strong year-end promotional push is necessary to prevent the seasonal slump and boost the 3.00% Repeat Rate.

- The first priority must be to protect the main revenue source by lowering the 96.81% churn rate for Credit Card users through targeted loyalty rewards. Since the data shows a direct link between payment length and retention, the business should push for more Long-term (13+) installment plans, as these users are over three times more likely to return than short-term buyers. To stabilize the Customer LTV, it is critical to investigate why the July 2017 cohort bottomed out at \$153.84 to ensure those mistakes don't happen again.
- To fix the performance in the Cohort Matrix, we should send vouchers to customers right after their first order to give them a reason to come back. Since Vouchers have the lowest churn at 93.87%, this "second-purchase" incentive would help bridge the gap for groups like May 2018 and turn one-time shoppers into repeat buyers, finally improving the low 1.03 purchase frequency.

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Product & Seller Performance



Insight

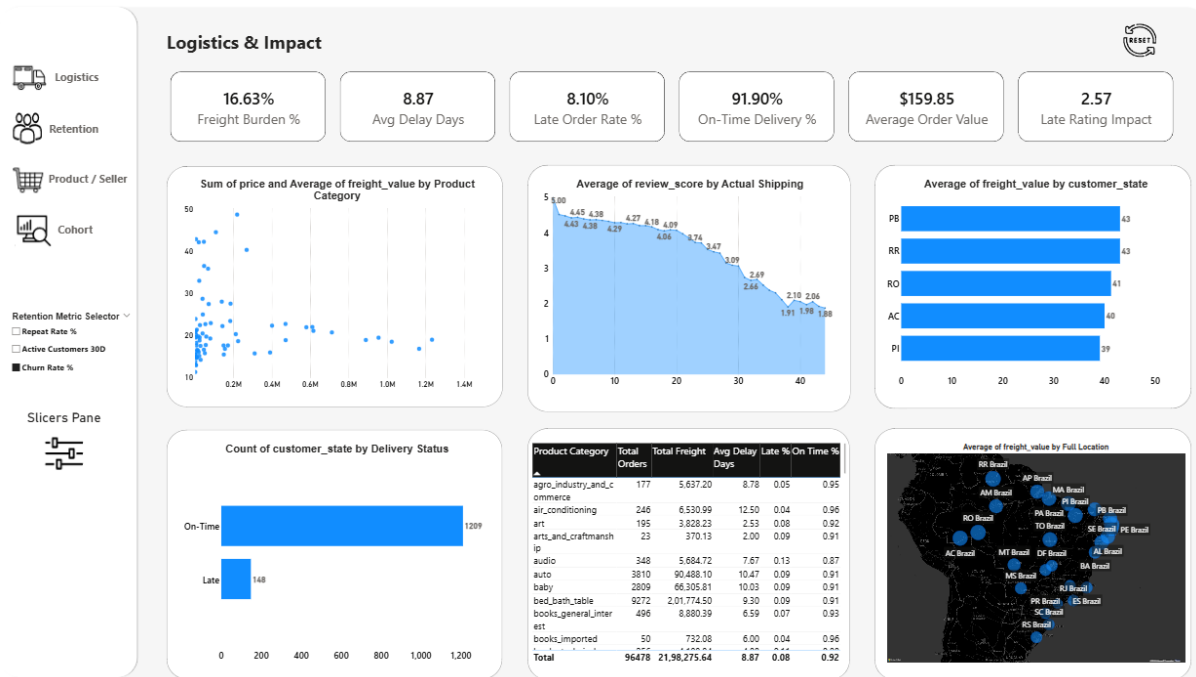
- Total results show \$15.42M in Total Revenue with an Average Review Score of 4.16. A major concern is that 5-star reviews have the lowest Repeat Rate at 3.78%, while 2-star and 3-star reviews actually show higher repeat rates of above 5%. Sales are heavily concentrated in the SP state, which accounts for 69.9% (\$10.02M) of all money made. While Health & Beauty is the top-selling category at \$1.42M, the Gaming PC category has a high 14.29% Cancellation Rate, which is the highest across all products.
- The scatter chart shows that while thousands of sellers struggle with low revenue and inconsistent ratings, a small group of elite sellers has successfully scaled to over \$200K while maintaining high 4.0+ review scores.

Recommendation

- The main priority should be growing sales outside State of SP, as depending on one state for nearly 70% of revenue is a risk. To improve quality, there needs to be an investigation into the Gaming PC category to fix the 14.29% cancellation rate so customers get their orders. It is also important to find out why 5-star customers are the least likely to return (3.78% repeat rate) and start a loyalty program to keep these happy customers coming back. Finally, since the top five sellers are each making over \$200K, These high-performing partners should be offered exclusive benefits or priority support to ensure they remain active and loyal among the 2,970 total sellers on the platform.
- Launch a Preferred Seller Program that provides exclusive marketing tools and faster support to these high-revenue, high-rating sellers to ensure they stay on the platform and continue driving the majority of sales.

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Logistics & Impact



Insight

- The logistics performance shows a 91.90% On-Time Delivery rate, meaning most customers get their orders when expected. However, 8.10% of orders are late, with an average delay of 8.87 days, which causes the average rating to drop by 2.57 points. the customer review score crashes from a high of 5.00 down to 1.88. Geographically, the Map and State Bar Chart show that northern states like PB and RR pay the highest shipping fees at \$43.
- The Scatter Chart (Sum of Price vs. Avg Freight) shows that for most products, shipping is very expensive compared to the item's price. Many dots are high up on the left side, meaning low cost items have high freight costs, which creates the 16.63% Freight Burden. The Delivery Status chart confirms there are 148 late orders that need attention to prevent poor reviews.

Recommendation

The first priority is to speed up deliveries so they don't exceed 20–30 days, as this is where the review scores drop most sharply. To lower the 16.63% Freight Burden, we should look for cheaper shipping partners in the North and Northeast regions where states like PB and RR are currently paying \$43. For the low-priced items shown in the Scatter Chart, the business should encourage "bundle buying" (buying more items at once) so the shipping cost is shared across more products. Finally, using the Product Category table and should focus on categories with high delay days to fix their specific delivery routes and protect the \$159.85 Average Order Value.

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Overall Recommendations

- Improve customer retention by focusing on converting one-time buyers into repeat customers through better post-purchase engagement
- Strengthen first-order experience by ensuring smooth delivery, better communication, and improved customer satisfaction
- Identify and retain high-value customers through targeted engagement and reactivation strategies
- Reduce customer drop-off after first purchase by introducing lifecycle-based campaigns (Day 0, Day 7, Day 30)
- Improve delivery consistency to ensure better customer experience and reduce negative impact on repeat behaviour
- Use cohort analysis regularly to track customer lifecycle performance and identify weak retention periods
- Reduce dependency on a small group of customers by improving engagement across broader customer segments
- Leverage product category insights to improve repeat purchase behaviour in low-retention categories
- Maintain stable payment systems while monitoring behavioural shifts in transaction preferences
- Use dashboards for continuous monitoring of retention, revenue distribution, and customer engagement trends

Conclusion

- This project highlights that the business is performing well in terms of revenue generation but faces a major challenge in customer retention and repeat purchase behaviour.
- The analysis shows that most customers make only one purchase, while a small group of high-value customers contributes a large portion of total revenue. Cohort analysis confirms that customer engagement drops significantly after the first purchase, indicating weak long-term retention.
- The study also shows that delivery performance, product categories, and customer experience play an important role in influencing repeat purchases. However, customer lifecycle engagement remains short across all segments.
- The dashboards provide a clear understanding of customer behaviour, revenue distribution, cohort patterns, product performance, and logistics impact, helping to identify key areas for improvement.
- Overall, this project supports better business decision-making by highlighting retention gaps, identifying high-value customers, and improving understanding of customer lifecycle behaviour for long-term growth.